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Predicting coronary artery disease with ensemble-based feature selection and grid search-tuned SVM under class imbalance conditions

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Abstract

Coronary artery disease (CAD) is a common heart condition that leads to numerous deaths. The standard diagnostic tool for CAD, coronary angiography, is expensive, time-consuming, and carries harmful side effects. To overcome these limitations, computer-aided decision-making through machine learning algorithms offers an effective alternative for early CAD detection. This study uses the Z-Alizadeh Sani dataset, comprising 303 samples with 54 features. Feature selection was performed using a combination of XGBoost and Random Forest algorithms to identify the most significant predictors. To mitigate class imbalance, three oversampling techniques were applied, including Conditional Tabular Generative Adversarial Network (CT-GAN), Adaptive Synthetic Sampling (ADASYN), and Synthetic Minority Over-sampling Technique (SMOTE). ADASYN outperformed CT-GAN and SMOTE in improving classification outcomes. Five classifiers including Multi-Layer Perceptron (MLP), Random Forest (RF), Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), and Support Vector Machine (SVM) were tested and optimized using both grid search and random search. The highest balanced accuracy was achieved by the ensemble feature selection combined with ADASYN oversampling and support vector machine (SVM) with grid search resulting in 96.86% and an AUC of 1.00. This approach highlights the potential for integration into clinical decision support systems to assist in early CAD diagnosis.

Keywords CAD prediction, Oversampling techniques, Feature selection, Grid search, Support vector machine

1 Introduction

In recent years, there has been a significant rise in Coronary Artery Disease (CAD) in developing countries, contributing to an increasing number of fatalities associated with this prevalent cardiovascular condition [1]. According to a report from the National Health and Nutrition Examination Survey (NHANES), the prevalence of CAD among adults over the age of 20 was observed to be 48% from 2013 to 2016, with rates rising as

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age increases [2]. Despite medical efforts aimed at preventing, diagnosing, and treating CAD, global mortality related to this disease continues to rise annually. In 2019, approximately 18.6 million deaths were attributed to CAD, and according to the World Health Organization (WHO), it accounted for about 32% of all deaths worldwide in 2020. Projections suggest that by 2030, this figure could reach around 23.6 million [1].

Several risk factors have been identified as contributors to CAD, including restricted blood flow due to fatty build-up in the coronary arteries, hypertension, oxygen deficiency in heart muscles, stress, diabetes, smoking, poor dietary habits, physical inactivity, and elevated cholesterol levels [3, 4]. The rise in mortality from heart attacks can also be attributed to a lack of patient awareness and medical knowledge, inadequate diagnostic tools, and a scarcity of clinical specialists [3].

Coronary angiography is a widely used diagnostic technique for CAD that accurately visualizes coronaries' intraluminal anatomy. However, it requires skilled operators and carries procedural risks and high costs, making it unsuitable as a routine screening tool for all patients. It is recommended only for those suspected of having the condition. Instead, Electrocardiography (ECG) and echocardiography are frequently utilized for clinical assessments and non-invasive testing, though they lack the sensitivity and specificity of coronary angiography [5].

Over the past decade, researchers have sought to enhance the sensitivity and specificity of non-invasive diagnostic methods by incorporating computer-aided decision-making and machine learning algorithms into CAD diagnosis [6–10]. These algorithms find application in various fields, including big data, social networks, the Internet of Things, computer-aided disease diagnosis, biomedical data analysis, cybersecurity, and intrusion detection [11].

The integration of metaheuristic optimization algorithms with machine learning has emerged as a promising approach for improving diagnostic accuracy across multiple medical domains. Recent advances in optimization-based medical diagnosis have shown significant improvements in classification performance. Elshewey et al. introduced the Greylag Goose Optimization (GGO) algorithm for enhancing heart disease classification, achieving a remarkable accuracy of 99.58% when combined with Long Short-Term Memory (LSTM) networks [12]. Their binary GGO (bGGO) algorithm effectively selected optimal features compared to six other binary optimization methods, demonstrating the potential of nature-inspired algorithms in cardiovascular diagnosis. Similarly, in the field of biomedical signal processing, the Modified Al-Biruni Earth Radius (MBER) algorithm has shown better performance in EEG-based eye state classification, achieving 96.12% accuracy with K-Nearest Neighbors when optimized for feature selection [13]. These developments highlight the growing importance of hybrid optimization approaches that combine multiple algorithmic strategies for enhanced medical diagnosis.

The application of optimization algorithms extends beyond cardiovascular diseases to other medical domains. Elshewey and Osman demonstrated the effectiveness of binary breadth-first search (BBFS) for orthopedic disease classification, achieving 99.41% accuracy when combined with Random Forest algorithms [14]. Their approach showed 47.29% lower average error compared to competing binary optimization methods, emphasizing the versatility of metaheuristic algorithms across different medical classification tasks.

Furthermore, In CAD diagnosis, a number of studies have employed machine learning techniques. For instance, Alizadehsani et al. utilized C4.5 decision tree classification and bagging algorithms on a dataset with 303 numerical samples to diagnose CAD [15]. Similarly, Hassannataj et al. focused on feature selection related to the disease using a ranking approach with a random tree model [16]. Arabasadi et al. also proposed both neural network and genetic algorithms individually and in combination on the same dataset [17].

The effectiveness of any diagnostic system is greatly influenced by the chosen algorithm and the number of features used in disease diagnosis and prediction. Selecting relevant features through specific algorithms can significantly enhance the testing performance of machine learning models when applied to unseen data samples in terms of accuracy and learning capacity [3].

High dimensionality is a prevailing challenge in machine learning, as model performance often deteriorates with an increasing number of features. This is due to increased model complexity and challenges in identifying optimal solutions. Furthermore, high-dimensional data can lead to overfitting, where the model closely aligns with the training data and fails to generalize well to new data.

As a result, feature selection is a common strategy employed to reduce dimensionality. By minimizing dimensions, one can decrease model complexity and improve generalization performance [18]. Within the context of CAD, various studies have targeted feature selection for dimensionality reduction. Typically, feature selection techniques are divided into three categories: filter methods, wrapper methods, and embedded methods.

Different studies have utilized these methods for feature selection in the Z-Alizadehsani dataset, including Information Gain and Gain Ratio [19], genetic algorithms [20], wrapper-based Boruta methods, and embedded approaches based on Support Vector Machines (SVM) [3], as well as recurrent feature elimination with Random Forest (RF) [2] and Whale Optimization Algorithm (WOA) [4].

Another notable issue with the Z-Alizadehsani dataset is class imbalance, which occurs when the number of observations in one class significantly outweighs those in another. To address this imbalance, Nasarian et al. implemented techniques such as the Synthetic Minority Oversampling Technique (SMOTE) and Adaptive Synthetic Sampling (ADASYN) [6]. Additionally, the SMOTE approach has been applied in studies [3, 4, 21] to mitigate this issue.

In the current study, five classification algorithms—Random Forest (RF), Logistic Regression (LR), Extreme Gradient Boosting (XGBoost), Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP)—were employed and systematically evaluated using both grid search and random search optimization strategies. To address the issue of class imbalance inherent in cardiovascular datasets, three oversampling techniques were conducted and comprehensively compared: SMOTE, ADASYN, and Conditional Tabular Generative Adversarial Network (CT-GAN). Results demonstrated that the ensemble feature selection combined with ADASYN oversampling and optimized SVM algorithm achieved exceptional performance with 96.67% accuracy, 100% precision and specificity, 95.24% sensitivity, and a perfect AUC score of 1.00, significantly outperforming CT-GAN (70.65% accuracy) and traditional approaches.

In summary, the primary contributions of this study are as follows:

- Implementing nested cross-validation with ensemble feature selection and oversampling performed independently within each fold.
- Developing a novel approach that combines Random Forest and XGBoost importance scores to identify the most clinically relevant features from cardiovascular risk factors, while reducing dimensionality while preserving critical diagnostic information.
- Evaluating three state-of-the-art oversampling techniques (SMOTE, ADASYN, CT-GAN) with comprehensive analysis revealing ADASYN's superiority through adaptive synthetic sample generation in challenging decision boundary regions.
- Conducting systematic hyperparameter optimization using both grid search and random search across five classification algorithms.
- Implementing SHAP (SHapley Additive exPlanations) analysis to provide transparent feature contribution explanations, with clinical interpretation revealing that Typical Chest Pain (SHAP = 0.567), Atypical chest pain (SHAP = 0.326), EF-TTE (SHAP = 0.204), Age (SHAP = 0.201), and Hypertension (SHAP = 0.179) are the most influential predictors for CAD diagnosis.
- Demonstrating model robustness across four publicly available cardiovascular datasets (Z-Alizadeh Sani, Cleveland, Statlog, Long Beach VA) to establish external validity and clinical applicability across diverse patient populations.
- Conducting comprehensive statistical tests including McNemar's tests and paired t-tests to provide robust evidence for the statistical significance of performance improvements over baseline methods.
- Achieving 96.86% balanced accuracy, 100% precision and specificity, 95.24% sensitivity, and AUC of 1.00 results with the Ensemble FS + ADASYN + grid search-SVM approach.
- Achieving an average training time of only 0.91 s, the proposed Ensemble FS + ADASYN + Grid Search-SVM model demonstrates outstanding computational efficiency.

The structure of this study is as follows: Sect. 2 reviews previous works on machine learning techniques for CAD diagnosis and positions our contributions within the current literature. Section 3 outlines our comprehensive proposed methodology including ensemble feature selection, class imbalance handling, and nested cross-validation framework. Section 4 presents the comprehensive results from our approach including performance metrics, statistical significance tests, multi-dataset validation, and SHAP-based clinical interpretations. Finally, Sect. 5 concludes the study, discusses limitations, and outlines future research directions including metaheuristic optimization, EHR integration, and federated learning approaches.

2 Related works

In earlier studies, multiple machine learning techniques were utilized, along with methods for tackling data imbalance, dimensionality reduction, and relevant feature selection, all focused on diagnosing CAD using the Z-Alizadehsani dataset, which contains 303 numerical entries.

Alizadehsani et al. employed C4.5 decision trees and bagging algorithms to diagnose CAD [15]. They applied information gain and Gini index methods for feature selection. Their study focused on analyzing coronary artery narrowing, particularly in the left

anterior descending (LAD), Left Circumflex (LCX), and Right Coronary Artery (RCA). The performance of these algorithms was assessed using a 10-fold cross-validation approach. Results showed that the bagging algorithm surpassed the C4.5 algorithm, with accuracy rates of 79.54%, 65.09%, and 66.31% for identifying stenosis in the LAD, LCX, and RCA, respectively, compared to 76.56%, 63.10%, and 63.38% for the C4.5 algorithm.

Arabasadi et al. explored the application of neural networks and genetic algorithms, both independently and in conjunction [17]. A key factor influencing the performance of the neural network was the selection of suitable initial weights. Random weight initialization led to increased output errors. For feature selection, they employed ranking methods such as weight by SVM, Gini index, information gain, and principal component analysis. Their results indicated that the accuracy rates for the neural network and the combined neural-genetic network algorithms, utilizing a 10-fold approach, were 84.62% and 93.85%, respectively. Furthermore, the use of SVM for feature selection enhanced neural network performance by approximately 10% through increased initial weights, and even more than 20% when coupled with a genetic algorithm on training data.

Alizadehsani et al. also presented a feature engineering strategy aimed at enhancing the accuracy of non-invasive CAD diagnosis through the results derived from Bayesian networks, C4.5, and SVM classifiers [7]. They expanded their dataset from 303 to 500 records and incorporated SVM weighting as a method for feature selection across all classification approaches. The accuracy rates achieved using the 10-fold cross-validation technique were 86% for Naïve Bayes, 89.8% for C4.5, and 96.40% for SVM algorithms.

In research conducted by Abdar et al., a combined two-level genetic algorithm and support vector machine, termed N2Genetic-NuSVM, was introduced for CAD diagnosis [10]. This method employs a two-level genetic algorithm aimed at simultaneously optimizing SVM parameters and feature selection. This approach improved CAD diagnosis accuracy to 93.08% when using a 10-fold cross-validation method.

Hassannataj Joloudari et al. conducted a feature selection study for CAD diagnosis utilizing a ranking method based on the random tree model [16]. They evaluated various models, including support vector machine, C5.0 decision trees, and CHAID, against the random tree model. Ultimately, the random tree model identified 40 features with a Gini index of 93.40%, achieving an Area Under the Curve (AUC) accuracy of 96.47% and an overall accuracy of 91.47%, surpassing the other classification models in these aspects.

Nasarian et al. introduced a novel hybrid feature selection algorithm tailored for CAD diagnosis, named heterogeneous hybrid feature selection [11]. This algorithm was tested on the Nasarian dataset, which encompasses clinical cases as well as workplace and environmental factors. Data imbalance was addressed using techniques such as SMOTE and ADASYN. Various classifiers, including decision trees, Gaussian Naïve Bayes, Random Forest (RF), and XGBoost, were utilized. The results showed that the proposed feature selection method, when implemented with SMOTE and XGBoost, achieved a classification accuracy of 81.23%. Additionally, accuracies of 83.94%, 81.58%, and 92.58% were attained for the Hungarian, Long-beach-va, and Z-Alizadeh Sani datasets, respectively.

In a study by Alizadehsani et al., a new feature selection method called assurance feature selection was proposed [5]. They addressed uncertainty in CAD prediction via data discretization and employed a genetic algorithm for determining SVM kernel hyperparameters. The achieved accuracy rates for diagnosing stenosis in major coronary arteries,

such as RCA (82.85%), LCX (83.47%), and LAD (86.64%), underscored the method's effectiveness.

Hassannataj Joloudari et al. introduced a hybrid machine learning framework known as Genetic Support Vector Machine and Variance Analysis (GSVMA) [20]. In this approach, analysis of variance (ANOVA) served as the kernel function for the SVM algorithm. The GSVMA model applied to the Z-Alizadeh Sani dataset benefited from a genetic optimization algorithm for feature selection. Using this method alongside SVM with ANOVA, Linear SVM (LSVM), and LIBSVM with Radial Basis Function (RBF), the results illustrated that GSVMA outmatched other methods, achieving the highest accuracy rate of 89.45% through a 10-fold cross-validation with 31 selected features.

Jin and Li developed a computer-aided diagnosis model for CAD based on a K-Nearest Neighbor (KNN) technique integrated with a Whale Optimization Algorithm (WOA) for feature selection, and a stacking model for diagnosis and prediction [2]. The WOA algorithm facilitated optimal continuous feature selection and binary transformation to derive the best feature subsets for each main coronary artery. Subsequently, a two-layer cumulative model was created based on these chosen subsets for the detection of LAD, LCX, and RCA. The proposed approach selected 17 features for each artery and achieved classification accuracies of 89.68%, 88.71%, and 85.81% across the respective test sets.

Suryani et al. incorporated an SVM-based model paired with feature selection techniques for CAD diagnosis [22]. Their feature selection technique represented a blend of wrapper and filter approaches, utilizing a genetic algorithm in the wrapper method alongside SVM. In the filtering step, they employed a Fast Correlation-Based Filter (FCBF), which followed an oversampling procedure utilizing ADASYN to combat data imbalance. Their deep neural network detection system focused on metrics such as accuracy, sensitivity, positive predictive value, and AUC. The test results revealed that their model, utilizing 5 features out of 54, attained an AUC of 93.7%, an accuracy of 87.7%, and a sensitivity of 87.7%.

Kolukisa and Bakir-Gungor investigated seven feature selection approaches, such as chi-square, gain ratio (GR), information gain, relief F, Support Vector Machine Feature Selection (SVM FS), Bee Search (BS), and conditional mutual information maximization, in addition to techniques informed by domain expertise and a range of classification algorithms [1]. Their work also introduced exhaustive and probabilistic ensemble methods for feature selection. The multilayer perceptron classifier achieved the highest accuracy rates, recording 91.78% on the Z-Alizadeh Sani dataset, 85.55% on the Statlog dataset, and 85.47% on the Cleveland dataset.

Khozeimeh et al. presented an Active Learning with Ensemble of Classifiers (ALEC) method for diagnosing CAD [23]. This strategy consists of four classifiers; three assess the presence of stenosis in the primary coronary arteries, while the fourth predicts CAD status. The ALEC technique was initially trained on labeled samples and subsequently evaluated on unlabeled samples. If the classifier outputs were consistent, these samples were incorporated into the labeled set. Medical experts manually labeled heterogeneous samples before inclusion. Compared to 19 alternative active learning algorithms, the combination of ALEC with the SVM classifier achieved impressive performance, with an accuracy of 97.01%.

Recent studies have demonstrated significant improvements in CAD diagnosis through advanced machine learning and optimization techniques. Elshewey et al. introduced the Greylag Goose Optimization (GGO) algorithm for enhancing heart disease classification accuracy [12]. The research utilized multiple classifiers, with Long Short-Term Memory (LSTM) networks emerging as the most effective, achieving 91.79% accuracy. When hyperparameters were optimized using GGO, the GGO-LSTM model achieved better performance with 99.58% accuracy. The statistical validation employed Wilcoxon signed-rank tests and ANOVA to assess both feature selection and classification outcomes, confirming the robustness of the hybrid approach. This approach has proven particularly valuable in medical applications where feature selection is crucial for achieving high diagnostic accuracy while maintaining computational efficiency.

Elshewey et al. presented the Modified Al-Biruni Earth Radius (MBER) algorithm for improving eye state classification precision using EEG signals, focusing on binary classification of eye states (open/closed) [13]. The MBER algorithm's binary format significantly enhanced feature selection capabilities compared to traditional optimization methods, including BER, PSO, WAO, GWO, and GA. Using comprehensive preprocessing techniques and statistical validation through ANOVA and Wilcoxon signed-rank tests, the K-Nearest Neighbors (KNN) classifier achieved optimal performance with 96.12% accuracy when optimized using MBER, demonstrating precision of 0.959425, NPV of 0.964969, and F-score of 0.963431.

Elshewey and Osman developed an innovative approach using binary breadth-first search (BBFS) for orthopedic disease classification, comparing it with BPSO, BGWO, and BWAO for feature selection on a dataset of 310 instances with six biomechanical features [14]. BBFS achieved a 47.29% lower average error compared to competing algorithms. Among six machine learning models evaluated (RF, SGD, NBC, DC, QDA, and ET), the Random Forest model demonstrated 91.4% accuracy before optimization, but when hyperparameters were optimized using various algorithms, the BFS-RF combination achieved exceptional performance with 99.41% accuracy, outperforming other hybrid models.

Tarek et al. utilize nature-inspired optimization algorithms for improving feature selection in cardiovascular disease prediction [24]. Their novel framework, called CVD-SO (Cardiovascular Disease-Snake Optimization), leverages snake optimization algorithms to address the challenge of high-dimensional medical data. Their approach integrates five different machine learning methods within a unified framework to efficiently select and classify relevant medical data. The remarkable achievement of 99.9% accuracy in cardiovascular disease detection demonstrates the effectiveness of bio-inspired optimization methods in medical diagnosis applications, representing a paradigm shift toward more sophisticated automated diagnostic systems that could substantially reduce CVD-related mortality rates through early intervention strategies.

Furthermore, the application of multitask learning architectures in critical care environments has emerged as a powerful approach for the simultaneous prediction of multiple clinical outcomes. For example, El-Rashidy et al. developed a comprehensive multitask multilayer-prediction model specifically designed for predicting mechanical ventilation requirements and associated mortality rates in intensive care units using the extensive MIMIC-III dataset [25]. Their innovative approach utilized data extracted across multiple time frames (6, 12, and 24 h) with multilayer multitask long short-term

memory (LSTM) models for classification tasks and neural networks with multilayer perceptron architecture for regression tasks. The study achieved promising results with accuracy rates of 0.944 for ventilation prediction and 0.971 for mortality prediction when utilizing 24-hour observation windows, proving particularly valuable for real-time patient management and hospital resource allocation.

While these recent advances highlight the importance of hybrid approaches in medical diagnosis, the current study addresses several critical gaps in existing CAD prediction methodologies through a comprehensive multi-faceted approach. Unlike previous works that often apply preprocessing techniques to entire datasets before cross-validation, our study implements a rigorous nested cross-validation framework where ensemble feature selection and oversampling are performed independently within each fold, preventing data leakage and ensuring unbiased performance estimates as recommended by established methodological guidelines. Our approach encompasses multiple dimensions that advance the field. First, we conduct three advanced oversampling techniques including SMOTE, ADASYN, and CT-GAN that are specifically optimized for the challenging class imbalance of Z-Alizadeh Sani dataset. Our analysis revealed ADASYN's superiority through adaptive synthetic sample generation achieving 96.86% balanced accuracy, 100% precision and specificity, 95.24% sensitivity, and perfect AUC of 1.00. Second, we develop an ensemble feature selection approach that combines Random Forest and XGBoost importance scores to identify the most clinically relevant features from the original 54 cardiovascular risk factors, providing superior dimensionality reduction while preserving critical diagnostic information. Third, our study systematically evaluates five classification algorithms (random forest, logistic regression, XGBoost, SVM, and MLP) using both grid search and random search optimization strategies. Fourth, we enhance clinical interpretability through SHAP analysis, providing transparent feature contribution explanations with clinical interpretation. Fifth, unlike studies limited to single-dataset validation, we demonstrate model robustness across four publicly available cardiovascular datasets (Z-Alizadeh Sani, Cleveland, Statlog, Long Beach VA) to establish external validity and clinical applicability across diverse patient populations. Sixth, we provide comprehensive statistical validation through McNemar's tests and paired t-tests to establish robust evidence for the statistical significance of performance improvements over baseline methods, addressing a critical gap in existing literature where statistical significance is often overlooked.

Finally, our approach maintains exceptional computational efficiency with training times of 0.91 s, making it highly suitable for real-world clinical deployment where both accuracy and speed are essential. Through this comprehensive evaluation framework, we demonstrate that the proposed Ensemble FS + ADASYN + SVM methodology achieves clinical-grade performance that significantly outperforms existing approaches while maintaining computational efficiency and clinical interpretability, thus bridging the gap between algorithmic sophistication and practical clinical deployment for early-stage CAD diagnosis. This multi-dimensional contribution establishes new benchmarks for non-invasive CAD prediction and provides a robust foundation for future research in cardiovascular machine learning applications.

3 The proposed methodology

In our study, to predict CAD, an optimized support vector machine based on the ensemble algorithm with the selection of important features was used on the Z-Alizadeh Sani dataset. The proposed methodology is shown in Fig. 1.

According to Fig. 1, the proposed methodology has been carried out in nine different phases. These phases include searching for studies on the prediction CAD (phase 1), data set description (phase 2), data preprocessing (phase 3), data set partition (phase 4), feature selection (phase 5), Data balancing (phase 6), Classification models (phase 7), optimization of hyperparameters (phase 8), model evaluation criteria and interpretation of results (phase 9). Then, after the completion of nine phases, a diagnosis and decision regarding CAD are made. The phases of the proposed methodology are described in detail. In addition, the phases related to the evaluation of models and results are explained in Sect. 4.

3.1 Searching for studies on the prediction of CAD

The first phase is to search for studies on predicting CAD using machine learning methods. These studies were searched and extracted through web search engines, including Google Scholar and PubMed.

3.2 Data set description

The dataset utilized in this study is the Z-Alizadeh Sani dataset, which can be accessed at the UCI database[1]. This particular dataset is one of the most common datasets used in machine learning for automatic CAD detection and comprises 303 samples, with 216 being CAD patients and 87 classified as normal patients, all having 54 features. Within this dataset are key features such as clinical features, signs and symptoms, echocardiography (ECG), and laboratory tests. Research variables consist of both dependent and independent variables. The dependent variable, labeled as cath, is defined with two classes: normal and CAD. Independent variables are considered to be the features found within the aforementioned dataset.

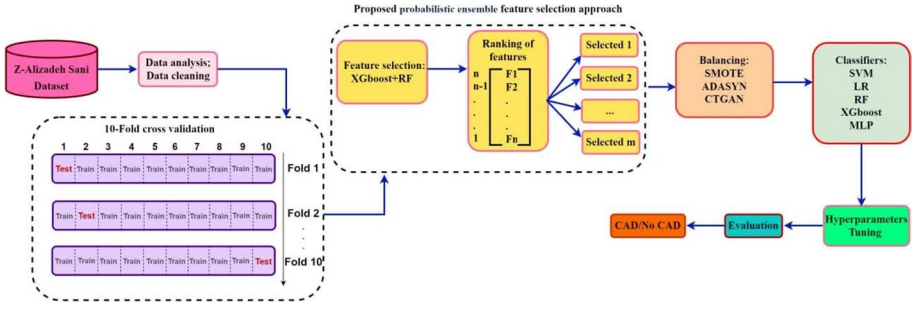


Fig. 1 The proposed methodology

[1] https://archive.ics.uci.edu/dataset/412/z+alizadeh+sani.

3.3 Data preprocessing

3.3.1 Statistical analysis

In the pre-processing stage, it was found that out of 55 features, 54 are independent and one is dependent. By examining the statistical analysis, the hidden relationships among the data were further understood. There are also 21 (quantitative) features.

To validate the statistical performance of our model we comprehensively compared statistical significance tests to compare the proposed methodology against baseline models. McNemar's test was employed to assess the statistical significance of differences in classification disagreements between paired models, with results reported as McNemar's χ^2 statistic and corresponding p-values. Additionally, paired t-tests were performed on ROC AUC scores obtained from 10-fold cross-validation to evaluate the statistical significance of performance differences, with t-test statistics and p-values calculated for each comparison. These statistical tests were conducted at $\alpha = 0.05$ significance level to provide robust evidence for the superiority of our proposed Ensemble FS + ADASYN + SVM approach over alternative models. The complete statistical analysis results, including McNemar's χ^2 values, McNemar p-values, paired t-test statistics for ROC AUC, and corresponding t-test p-values, are presented in the Results and Discussion section.

3.3.2 Handling missing data

By handling the status of missing data with the help of the Pandas library, it was found that there is no missing data in this data set.

3.3.3 Normalization

In the normalization part, the data must be converted from a string to numerical values. Also, during the normalization process, the dataset was scaled to ensure that quantitative variables were transformed into values ranging from 0 to 1. The RobustScaler method was employed for this specific task [26]. The most common formulation is presented as follows:

$$X_{scaled} = \frac{X - Median\,(X)}{IQR\,(X)} \tag{1}$$

where X_{scaled} represents the normalized output value indicating how many interquartile ranges the original data point deviates from the median. X is the original input feature value from the dataset requiring transformation, Median(X) serves as the middle value of the feature distribution, and IQR(X), the Interquartile Range calculated as the difference between the 75th and 25th percentiles, represents the spread of the middle 50% of data.

This method, known as robust statistics, aims to normalize data by centering and scaling features using the median and IQR. Unlike StandardScaler, which standardizes data by subtracting the mean and dividing by the standard deviation, RobustScaler is less sensitive to outliers, making it an ideal choice for datasets with extreme values or skewed distributions. This process starts with the centrality of each feature by subtracting its median, effectively reducing the central tendency of the data to zero. Then, each feature is scaled by dividing it by the IQR. By focusing on the median and IQR, RobustScaler preserves the underlying shape of the distribution while minimizing the impact of

outliers. This is especially valuable for data sets that do not follow a normal distribution and ensures that data normalization does not distort the underlying characteristics of the data. Overall, RobustScaler provides a robust and flexible method for data normalization in research environments where outliers and non-normal distributions exist.

3.4 Data set partition

Stratified k-fold cross-validation was used to divide the data set into training and test data, considering 10 folds. This approach avoids overfitting. The use of stratified K-fold cross-validation is beneficial in scenarios involving unbalanced data and also in cases where it is necessary to preserve the class distribution among distinct folds [26]. To prevent data leakage and ensure robust model evaluation, all preprocessing steps, including feature selection and oversampling, were performed independently within each cross-validation fold. For each of the 10 stratified cross-validation folds, the dataset was divided into training (90%) and testing (10%) subsets while maintaining class distribution, after which the ensemble feature selection process combining Random Forest and XGBoost importance scores was applied exclusively to the training portion of each fold. The selected features were then applied to the corresponding test set using the same feature indices identified from the training data, and oversampling techniques (SMOTE, ADASYN, and CT-GAN) were applied only to the training data within each fold, ensuring no synthetic samples influenced the test set.

3.5 Feature selection

Feature selection was performed inside each fold to avoid bias, consistent with prior studies that demonstrate the importance of preventing information leakage during preprocessing. Krawczuk and Łukaszuk demonstrated that using the same dataset for feature selection and model evaluation can introduce significant bias ranging from 2.6% to 41.67% [27]. They also strongly recommended cross-validation for feature selection to reduce selection bias. Similarly, Abeel et al. highlighted the use of robust feature selection methodologies in biomedical applications [28]. To ensure stability and prevent overfitting in high-dimensional datasets, they advocated for ensemble approaches and proper validation frameworks. Güney further demonstrated that feature selection should be performed exclusively on training data to avoid bias [29]. Additionally, Güney addressed the critical interdependency problem between feature selection and classifier optimization, introducing a feature selection-integrated classifier optimization algorithm that incorporates feature selection during classifier optimization to enhance performance while reducing computational complexity [30]. This approach achieved superior accuracies across multiple datasets while demonstrating that integrated optimization algorithms can improve classifier accuracy with fewer features, directly supporting our methodology's approach of combining ensemble feature selection with optimized SVM classification.

Consistent with these recommendations, our ensemble feature selection approach combined Random Forest and XGBoost importance scores, and it is applied independently within each cross-validation fold, ensuring that no information from test sets influences the feature selection process. This nested cross-validation framework prevents the feature selection bias problem while maintaining the robustness of our

42-feature selection from the original 54 cardiovascular risk factors, thus providing unbiased performance estimates that reflect true model generalization capability.

The feature selection process is as follows:

1. *Model training:* First, a random forest model and an XGBoost model are simultaneously trained on a dataset containing the feature matrix (x) and the target variable (y).
2. *Importance score extraction:* After training, feature importance scores are extracted from each model. Random Forest uses the feature_importances_ attribute, while XGBoost provides feature_importances_ which indicates the importance of the feature in the gradient boosting process.
3. *Merging scores:* The essential nature of the hybrid technique lies in combining feature importance scores from both models. Integration with the mean importance scores is obtained from Random Forest and XGBoost, with the aim of balancing the biases of each model for a more stable and flexible assessment of feature importance.
4. *Attribute ranking:* After calculating the combined importance scores, the attributes are ranked based on these scores. The ranking is done by sorting the index scores in descending order and creating a list of attributes ranked by relative importance.
5. *Using ranked features:* A ranked list of features can serve various purposes, including reducing the feature set to increase computational efficiency, emphasizing the most relevant features to increase model performance, or gaining insight into the most influential features in predicting the target variable.

Generally, a combination of feature importance scores from Random Forest and XGBoost has several advantages.

(1) *Bias reduction:*By averaging the scores of different models, this technique can reduce the biases of individual models and provide a fairer assessment of feature importance.
(2) *Improved robustness:* Using multiple models to assess feature importance produces a more reliable estimate and reduces the likelihood of overfitting with a particular feature selection strategy.
(3) *Adaptability:* The method can be adjusted for diverse datasets and machine learning scenarios, providing a versatile approach to feature selection.

3.6 Data balancing

Considering the imbalance present in the Z-Alizadeh Sani dataset, applying data balancing techniques is essential to ensure fair learning across classes. Among the various approaches available, this study employed three synthetic oversampling methods—SMOTE, ADASYN, and CT-GAN—to mitigate bias towards the majority class and improve the generalization capability of the models.

SMOTE is a widely used method that works by generating synthetic examples along the line segments joining minority class instances and their nearest neighbors [31]. Instead of simply duplicating existing minority samples, SMOTE introduces slight variations, which helps reduce the risk of overfitting and creates a smoother decision boundary between classes.

ADASYN builds on SMOTE by focusing more on the minority samples that are harder to learn—those lying near class boundaries or surrounded by majority instances [32]. By adaptively generating more synthetic data for these complex regions, ADASYN allows

the classifier to better focus on difficult examples and typically results in more balanced classification performance.

CT-GAN, on the other hand, represents a deep learning-based strategy designed specifically for generating synthetic tabular data [33]. Unlike traditional oversampling techniques, CT-GAN uses adversarial training to learn the conditional distribution of minority class data and generate realistic synthetic samples conditioned on class labels.

The enhanced performance of ADASYN over CT-GAN (96.67% vs. 70.65% accuracy) can be attributed to several fundamental differences in their data augmentation approaches. ADASYN's adaptive density-based sampling generates synthetic samples proportionally to the difficulty of learning each minority class instance, focusing on regions near decision boundaries where classification is most challenging, which is particularly effective for the CAD dataset where minority class patients exhibit complex patterns near the classification boundary. In contrast, CT-GAN's limitations became apparent due to insufficient training data, as GANs typically require thousands of samples to learn complex data distributions accurately, while our dataset contains only 303 samples. Additionally, CT-GAN struggled with the heterogeneous nature of medical features and training instability inherent in GANs, likely resulting in mode collapse where the generator produces limited diversity in synthetic samples, leading to the observed lower specificity (99.09%) compared to ADASYN's perfect specificity (100%). The performance differences highlight that for medical diagnosis tasks with limited data, data quality enhancement through intelligent oversampling (ADASYN) is more effective than data quantity increase through generative approaches (CT-GAN). While ADASYN preserves the original data distribution and generates clinically plausible synthetic samples, CT-GAN's architecture, despite being designed for tabular data, cannot adequately capture subtle correlations between cardiovascular risk factors with the available sample size.

In our workflow, these oversampling methods were applied after splitting the dataset into training and testing subsets. This ensured that only the training set was artificially balanced, avoiding data leakage.

3.7 Classification models

In this study, 5 classification algorithms, such as RF, logistic regression, XGBoost, and SVM, and MLP were used.

1) Random forest.

Random Forest (RF) is a widely used algorithm in supervised machine learning [34]. This method can effectively handle both classification and regression tasks. It operates on the principle of ensemble learning, which involves combining multiple classifiers to tackle complex problems and enhance model performance. As its name implies, "Random Forest is a classifier made up of several decision trees built on various subsets of a given dataset, which are averaged to increase the accuracy of predictions." Unlike relying on a single decision tree, RF gathers predictions from each tree and uses majority voting to determine the final output. A greater number of trees in the ensemble tend to improve accuracy while mitigating the risk of overfitting.

The Random Forest algorithm functions in two main phases. The first phase involves creating the random forest by assembling N decision trees, and the second phase entails making predictions based on these generated trees.

The process can be outlined in the following steps: Step 1: Randomly select K data points from the training dataset. Step 2: Build decision trees based on the selected data point subsets. Step 3: Decide on the number N of decision trees to construct. Step 4: Repeat steps 1 and 2 as needed.

2) Logistic regression.

Logistic regression is a machine learning method categorized under supervised learning techniques [22]. It is primarily employed to forecast a dependent variable based on a specific set of independent variables. This model is particularly useful when the dependent variable is categorical.

In logistic regression, rather than fitting a traditional regression line, an "S"-shaped logistic function is applied, which predicts two possible outcomes: 0 or 1. Essentially, logistic regression operates by establishing a threshold value that determines the probability of the outcomes being either 0 or 1. Values exceeding this threshold are classified as 1, while those below it are classified as 0.

To derive the logistic regression equation, one can start from the linear regression equation. The process to obtain the logistic regression equations involves several mathematical steps, beginning with the formulation of the linear equation.

$$y = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \ldots + b_nx_n \tag{2}$$

In logistic regression, y can only be between 0 and 1, so the above equation is divided by (1-y).

$$\frac{y}{1-y}; \tag{3}$$

0 for y=0, and infinity for y=1.

But there is a need for a range between -[infinity] to +[infinity], then the logarithm of the equation is obtained.

$$log\left[\frac{y}{1-y}\right] = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \ldots + b_nx_n \tag{4}$$

The above equation is the final equation of logistic regression.

3) XGBoost.

XGBoost, which stands for "Extreme Gradient Boosting," has gained significant popularity as an algorithm owing to its effectiveness in managing large datasets and delivering high performance across various machine learning tasks like classification and regression. The field of machine learning has matured, leading to its implementation in numerous real-world applications, ranging from financial data analysis to medical forecasting. Notably, the XGBoost algorithm is an ensemble learning technique that merges the predictions from multiple weaker models to generate a more robust prediction [35].

One of the standout features of XGBoost is its proficiency in managing missing values, enabling it to work with real-world datasets containing incomplete information without

the need for extensive preprocessing. Furthermore, XGBoost is designed with built-in parallel processing capabilities, allowing for the training of models on large datasets within a reasonable timeframe.

4) Support vector machine

Support Vector Machine (SVM) is a widely utilized supervised learning algorithm applicable to both classification and regression tasks [16]. The primary objective of the SVM algorithm is to establish an optimal decision boundary that separates an n-dimensional space into distinct classes, enabling accurate classification of new data points. This optimal decision boundary is referred to as the hyperplane. The data points that are nearest to this hyperplane and influence its position are known as support vectors, hence the na.

SVM can be classified into two categories: linear and non-linear.

- *Linear SVM:* This variant is employed for datasets that are linearly separable, meaning that the data can be divided into two classes with a straight line. In this case, the classifier is characterized as a linear SVM classifier.
- *Non-linear SVM:* On the other hand, non-linear SVM is used for datasets that cannot be effectively classified with a straight line. Such datasets are termed non-linear, and the corresponding classifier utilized is the non-linear SVM.

5) Multi-layer perceptron

Multi-layer perceptron is like a versatile puzzle-solver in the world of machine learning, capable of untangling complex patterns in data to make predictions or classifications [36]. It's a type of artificial neural network that mimics how a brain might process information, using layers of interconnected nodes, or neurons, to transform raw inputs into meaningful outputs. Imagine feeding it a jumble of numbers, images, or text—like the pixel values of a photo or financial data—and watching it churn out answers, such as identifying a handwritten digit or forecasting tomorrow's stock prices. MLPs excel at these tasks by learning from examples, adjusting themselves over time to get better without needing explicit rules, making them a powerful tool for everything from speech recognition to medical diagnosis.

The journey of data through an MLP starts at the input layer, which acts like an open door, accepting raw information [37]. For an image, this might mean each pixel gets its own node; for a dataset, it could be numerical features like house sizes or ages. This data flows into one or more hidden layers, where the real computation happens. Each neuron in these layers takes inputs from the previous layer, applies a weighted sum, and passes the result through an activation function to decide what to send forward. These weights, which control the strength of connections between neurons, are the knobs the MLP tweaks to improve its accuracy. The deeper the network—with more hidden layers—the more complex the patterns it can learn, though this comes at the cost of needing more data and computational power. Finally, the output layer delivers the result, like a probability score for whether an email is spam or a predicted value for a house's price.

Learning is where the MLP truly shines, mimicking a trial-and-error process. It starts with random guesses, comparing its outputs to the correct answers using a loss function to measure how far off it is. Through a method called backpropagation, the network works backward, adjusting weights to reduce errors. This happens over multiple rounds, or epochs, gradually sharpening its predictions [38]. The activation function

plays a starring role here, deciding whether a neuron's output is significant enough to pass along. The sigmoid function, for instance, compresses outputs into a 0-to-1 range, ideal for probabilities, while ReLU (Rectified Linear Unit) outputs positive values or zero, speeding up training by avoiding certain pitfalls. These functions give MLPs the flexibility to model intricate relationships, whether it's spotting a cat in a photo or predicting market trends.

What makes MLPs so valuable is their adaptability. They can tackle diverse tasks—classifying objects in images, analyzing text for sentiment, or forecasting financial risks—without needing much prior knowledge about the data's structure. Their fully connected nature, where every neuron links to every neuron in the next layer, lets them uncover patterns in raw data. But they're not perfect. Training can be computationally intensive, especially for deep networks or large datasets, and MLPs often need substantial data to perform well [39]. They can also overfit, memorizing training data instead of learning general patterns. To counter this, techniques like dropout, which randomly ignores neurons during training, or regularization, which keeps weights in check, are often used to ensure robust learning.

Training an MLP involves splitting a dataset into training, validation, and test sets. The training set teaches the network, the validation set fine-tunes its performance, and the test set evaluates its final accuracy. An optimizer, like gradient descent, adjusts weights efficiently, guided by a loss function such as mean squared error for numerical predictions or cross-entropy for classifications. Over time, the MLP refines its weights to deliver better results. In practice, MLPs power applications like diagnosing diseases from medical data or recognizing speech patterns in audio [40]. Despite being a relatively simple neural network model compared to newer architectures, MLPs remain a go-to choice for their flexibility and effectiveness, proving that a well-trained, straightforward system can still deliver impressive results across a wide range of real-world challenges.

3.8 Optimization of hyperparameters

In the field of machine learning, hyperparameters play a vital role because they control the learning process and have a significant impact on model performance. Unlike model parameters, which are obtained through data-driven learning, hyperparameters are predetermined before the training step. Various search methods can be used to determine the optimal set of hyperparameters, below are detailed reviews of both grid search and random search methods.

Grid search, a conventional and straightforward approach to hyperparameter optimization, requires the creation of a hyperparameter grid followed by a methodical evaluation of all possible combinations in said grid. This method is characterized by its systematic nature, which allows comprehensive exploration in a predefined space [41]. Unlike grid search, random search provides a more consistent and sometimes effective alternative [42]. Rather than exhaustively examining every conceivable combination, random search selects random sets of hyperparameters within predefined constraints.

Both mentioned approaches have their advantages in hyperparameter optimization. Grid search is ideal when the hyperparameter space is relatively small or you want to perform an exhaustive search. On the other hand, random search is more flexible and efficient for larger spaces or when optimal meta-parameters are less predictable. In practice, the choice between the two approaches depends on the complexity of the model,

the dimensions of the hyperparameter space, and the computational resources. In our study, the classification models include: Random Forest, Logistic Regression, XGBoost, Support Vector Machine, and MLP. These were optimized once based on grid search and another time based on random search to comprehensively explore performance trade-offs. The settings of the mentioned models based on random search and grid search are presented in Table 1.

4 Experimental results

In this section, first, the implementation environment of the models is described. Then, the visualization based on the subplot is shown in Fig. 2. Next, the bar chart related to the statistical distribution of features is presented in Fig. 3. After that, the histogram diagram related to quantitative features is shown in Fig. 4. After that, the implementation environment of the models is described. Finally, the evaluation of the used models and the obtained results are interpreted in detail. Evaluation of models is done using criteria such as precision, accuracy, sensitivity, specificity, F1 score, and the AUC, so a matrix called a confusion matrix is used to obtain these criteria. The test results are related to the values of the above criteria, which are obtained by applying the confusion matrix elements to each of the stated criteria.

4.1 Implementation environment

The features related to the implementation environment of the models are as follows.

- Platform: Colab.
- Programming language: Python.
- Amount of external memory: 12 GB.
- Processor size: 64 bits.
- Platform: Windows 10.

Subletting a data set with CAD and normal refers to dividing the data into two separate categories based on the presence or absence of CAD.

By creating a subplot for subjects with CAD and those with normal heart function in the data set, researchers can compare and analyze the differences between subjects with

Table 1 Hyperparameter settings for models based on random and grid search

Model	Optimization method	Best settings
Random forest	Grid search	criterion='entropy', max_features='sqrt', min_samples_leaf = 1, min_samples_split = 2, n_estimators = 50
	Random search	criterion='gini', max_features='log2', min_samples_leaf = 2, min_samples_split = 4, n_estimators = 80
Logistic regression	Grid search	C = 1.0, penalty='l2', solver='liblinear', tol = 0.0001
	Random search	C = 0.1, penalty='l2', tol = 0.0001, solver='liblinear'
XGBoost	Grid search	max_depth = 5, learning_rate = 0.2, n_estimators = 150, objective='binary: logistic'
	Random search	max_depth = 3, learning_rate = 0.1, n_estimators = 100, objective='binary: logistic'
SVM	Grid search	C = 0.3, kernel='linear', probability = True
	Random search	C = 1.0, kernel='rbf', gamma='scale', probability = True
MLP	Grid search	hidden_layer_sizes=(100,), activation='relu', solver='adam', alpha = 0.0001
	Random search	hidden_layer_sizes=[(50,), (100,), (100, 50)], activation=['relu', 'tanh'], solver=['adam', 'sgd'], alpha=[0.0001, 0.001]

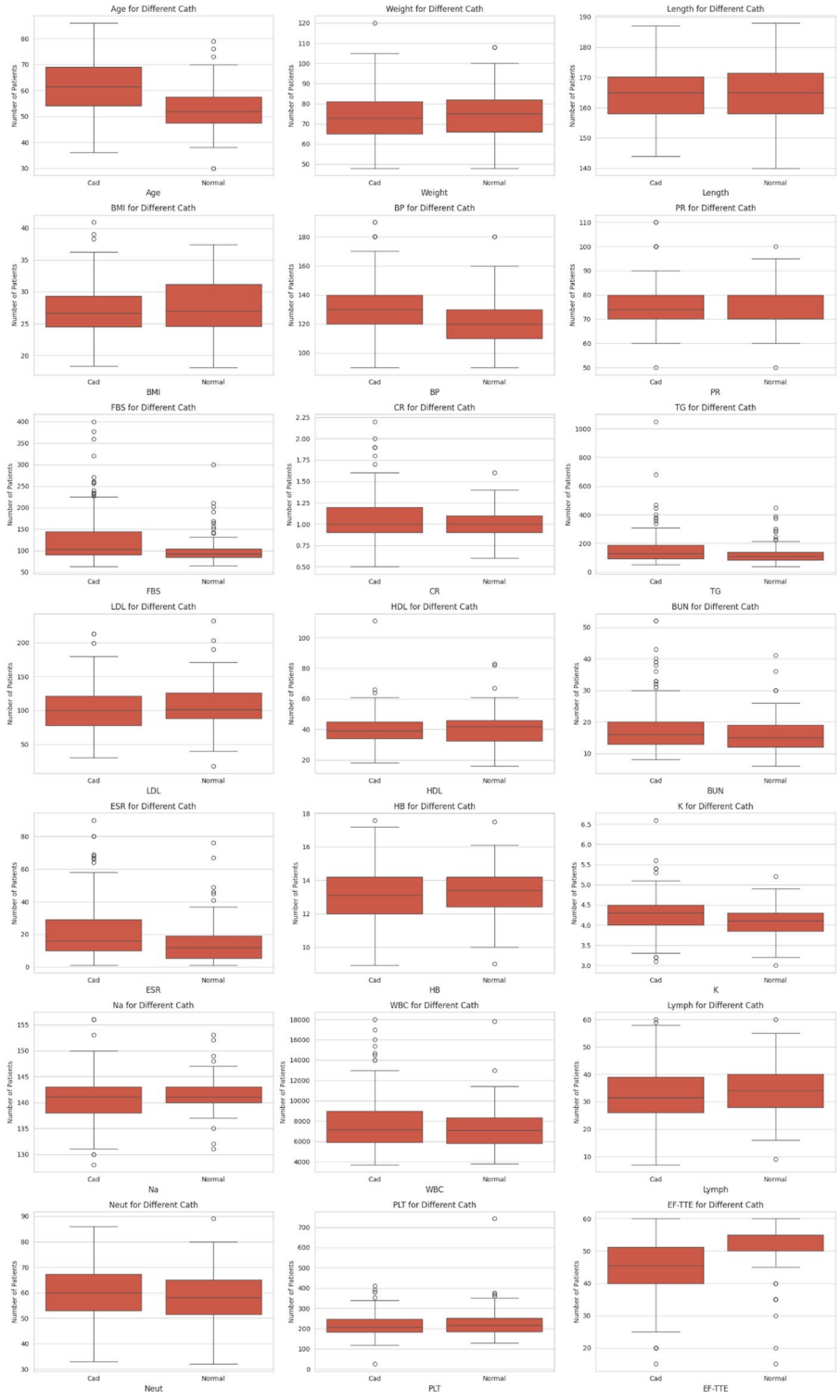


Fig. 2 Visualization based on subplot

CAD and those with normal heart function. This can help identify patterns, risk factors, and potential predictors of CAD.

Bar chart distribution of classified features in the CAD dataset such as gender, ex-smoker, HTN (high blood pressure), DM (diabetes mellitus), and CVA (cerebrovascular



Fig. 3 Bar chart related to the statistical distribution of features

accident), etc. provides a visual representation of the frequency or distribution of each category. For qualitative attributes such as gender, a bar chart shows the number of male and female subjects in the dataset with CAD. This can help to understand the gender distribution among CAD patients and identify any potential differences in disease prevalence between men and women.

Similarly, for characteristics such as ex-smoker, HTN, DM, and CVA, the bar chart illustrates the proportion of CAD patients who are ex-smokers, have hypertension, diabetes, and a history of cerebrovascular events, respectively. This could provide insights into the prevalence of these risk factors among individuals with CAD.

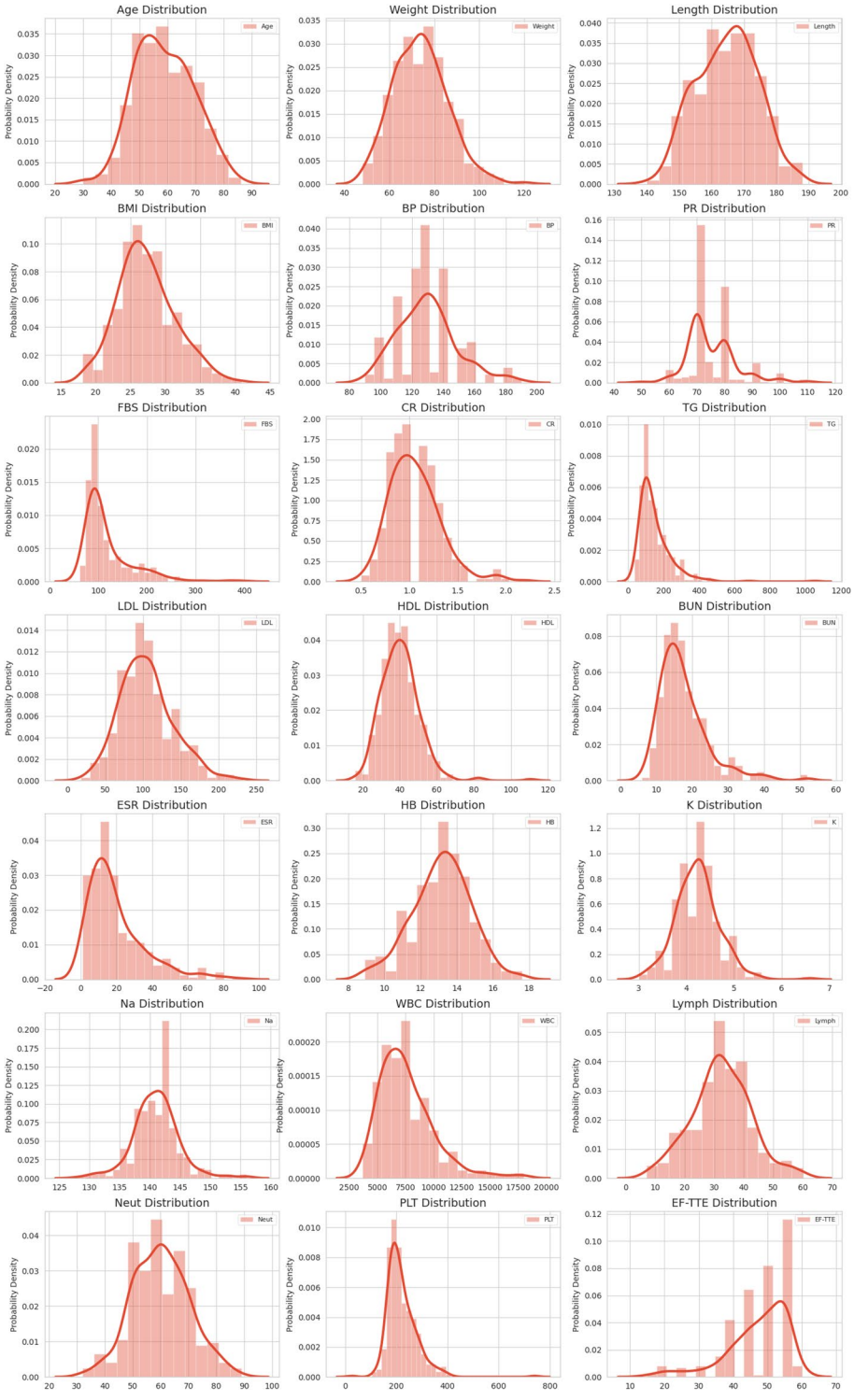


Fig. 4 Histogram chart related to quantitative features

Therefore, bar chart distribution analysis of categorical features can help researchers identify common features or risk factors associated with CAD in datasets. It can also help identify any patterns or correlations between these categorical variables and the presence of CAD.

Histograms of quantitative features such as age, body mass index, pulse rate, and other related variables play an important role in predicting CAD by providing insight into the distribution and characteristics of these factors in the dataset. These histograms provide a visual representation of the frequency or density of values within specific ranges, allowing researchers to identify patterns, outliers, and potential relationships with the presence or severity of CAD.

4.2 Evaluation criteria

In the seventh phase of the suggested methodology, the evaluation metrics for the models—such as precision, accuracy, specificity, F1 score, and the area under the curve—were computed using the confusion matrix. Essentially, this matrix serves as the basis for assessing the models' performance and accuracy. It comprises components like true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). The specifics of these matrix elements are outlined in Table 2, as referenced in [43].

According to the details provided in Table 2, the elements of the confusion matrix are explained as follows:

TP (True Positive): The number of positive cases correctly detected as sick by the test.

FP (False Positive): The number of positive cases wrongly classified as healthy by the test.

TN (True Negative): The number of negative cases properly recognized as healthy by the test.

FN (False Negative): The number of negative cases mistakenly labeled as sick by the test.

The confusion matrix plays a critical role in assessing and analyzing the effectiveness of classification techniques in distinguishing data across various groups. When data is divided into M groups, the classification matrix takes the form of at least an M x M grid. In an ideal scenario, most data points would align along the matrix's main diagonal, reflecting the TP and TN values, with the remaining off-diagonal entries being zero or near zero [44].

Additionally, the metrics used to evaluate the models are described below.

1) Accuracy

The accuracy criterion is the ability of a model to correctly predict healthy samples from diseased samples on the test data set. By observing the level of accuracy in the output of the model, it can be seen whether the model is properly trained or not, and that the overall performance of the model is examined.

The accuracy of a test is calculated as the ratio of the total number of correct positives and correct negatives to the total number of tested items. The formula for accuracy is as follows.

$$\mathrm{Acc} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{FP} + \mathrm{FN} + \mathrm{TP} + \mathrm{TN}} \tag{5}$$

Table 2 Confusion matrix

Actual class	**Predicted class**	
	Positive	**Negative**
Positive	TP	FP
Negative	FN	TN

2) Precision

Precision is a measure that determines how well the relevant model has performed in diagnosing sick people. The formula for correctness is described in relation (5).

$$\mathrm{PPV} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{6}$$

3) Specificity

The ability of a model to identify healthy people is measured by the specificity criterion. This criterion is defined as follows:

$$\mathrm{SPC} = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP}} \tag{7}$$

4) F1-score

F1 score is a suitable measure to evaluate the accuracy of a test. This measure considers precision and sensitivity together. This criterion is calculated as follows.

$$\mathrm{F1-Score} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}} \tag{8}$$

5) Area under the curve

The Area Under the Curve (AUC) criterion represents the area beneath the Receiver Operating Characteristic (ROC) curve. A higher AUC value for a model indicates better overall performance and more favorable evaluation outcomes. ROC curves are two-dimensional representations, with the vertical Y-axis displaying the true positive rate of the positive class and the horizontal X-axis showing the false positive rate of the negative class. Essentially, the ROC curve illustrates the trade-off and relative balance between the advantages and disadvantages associated with the classification results.

6) Sensitivity

Sensitivity (Recall), a critical metric in medical diagnosis, measures the model's ability to correctly identify positive cases (patients with CAD) [45].

$$Sensitivity = \frac{TP + FN}{TP} \tag{9}$$

7) Matthews Correlation Coefficient (MCC).

TP, TN, FP, FN are the confusion-matrix. MCC ranges from − 1 (total disagreement) through 0 (no better than chance) to + 1 (perfect prediction) [46].

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{10}$$

8) Per-fold training time (time complexity).

To calculate practical "time complexity" we measure the average wall-clock training time per cross-validation fold [47]. Where T_i is the elapsed training time for fold i (from start of preprocessing to completion of model fitting). k is the number of folds (e.g., k =

10). The resulting T_avg (in seconds) is recorded in Table 6 to indicate computational cost on the reference hardware.

$$T_{avg} = \frac{\sum_{i=1}^{k} T_i}{k} \tag{11}$$

9) Balanced accuracy.

Balanced accuracy provides a more appropriate measure for imbalanced datasets by equally weighting the performance on both classes. Values range from 0 to 1, where 1 indicates perfect classification [45].

$$Balanced\ Accuracy = \frac{Sensitivity + Specificity}{2} \tag{12}$$

10) Brier

Where n is the number of samples, p_i is the predicted probability for sample i, and y_i is the actual binary outcome for sample i (0 or 1). Lower Brier scores indicate better calibration, with 0 representing perfect calibration [45].

$$Brier = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2 \tag{13}$$

4.3 Results and discussion

During the model optimization process, it was observed that both grid search and random search methods perform similarly in identifying optimal hyperparameters. However, random search tends to be faster in some cases. Despite this, the SVM model optimized with grid search achieved impressive results, reaching an accuracy of 84.78%.

Among all the models evaluated, the SVM model with ensemble feature selection, SMOTE, and random search stood out with 85.77% accuracy, 75.39% precision, and 87.94% specificity, showcasing its outstanding performance. Even more notable was the SVM model using ADASYN and grid search, which achieved 96.67% accuracy, 100% precision, and 100% specificity, making it a competitive model in this study.

The detailed performance metrics—including accuracy, specificity, F1-score, AUC, balanced accuracy, brier, mcc and precision—of all the models used are summarized in Table 3 for Grid Search and Table 4 for Random Search. All machine learning models were optimized using grid search with 10-fold cross-validation. This approach ensures a systematic and exhaustive search for optimal hyperparameters, providing a fair basis for performance comparison.

The comparative analysis in Table 3 groups highlights performance differences among the tested models, with the Ensemble FS + ADASYN + SVM using Grid Search approach emerging as the most effective solution for CAD prediction. Traditional models as shown in Table 3 demonstrate varying performance levels when applied to the imbalanced CAD dataset. Random Forest (83.86% accuracy with grid search, 86.77% with random search) showed moderate accuracy but remained prone to overfitting due to noise from irrelevant features. Logistic Regression achieved competitive performance among traditional models (87.77% with grid search, 83.48% with random search) but was still constrained by its linear assumptions, which overlook complex feature interactions.

Table 3 Performance metrics with grid search

Models	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC	Balanced accuracy	MCC	Brier	Time complexity
Ensemble FS + SMOTE + Random Forest	83.86%	73.47%	88.01%	72.83%	72.17%	0.8888	80.74%	0.6175	0.1160	1.54 s
Ensemble FS + SMOTE + Logistic Regression	87.77%	90.97%	86.49%	75.09%	81.41%	0.9278	88.73%	0.7420	0.1074	1.89 s
Ensemble FS + SMOTE + XGBoost	84.18%	74.58%	88.01%	74.54%	73.60%	0.8928	81.30%	0.6325	0.1344	1.71 s
Ensemble FS + SMOTE + SVM	84.78%	87.08%	83.72%	70.10%	76.95%	0.9230	85.40%	0.6755	0.1127	1.51 s
Ensemble FS + SMOTE + MLP	87.09%	82.64%	88.87%	76.06%	78.78%	0.9138	85.76%	0.7012	0.1037	2.31 s
Ensemble FS + ADASYN + Random Forest	83.85%	78.06%	86.15%	71.87%	74.05%	0.8910	82.10%	0.6339	0.1194	3.35 s
Ensemble FS + ADASYN + Logistic Regression	83.82%	92.08%	80.50%	66.61%	76.84%	0.9252	86.29%	0.6755	0.1210	5.89 s
Ensemble FS + ADASYN + XGBoost	85.15%	80.28%	87.06%	74.49%	76.41%	0.8991	83.67%	0.6662	0.1231	3.07 s
Ensemble FS +ADASYN + MLP	84.78%	83.75%	85.17%	69.97%	75.88%	0.9012	84.46%	0.6589	0.1229	5.38 s
Ensemble FS + CT-GAN + Random Forest	58.82%	8.19%	79.26%	14.17%	9.70%	0.5096	43.73%	−0.1426	0.2423	6.27 s
Ensemble FS + CT-GAN + Logistic Regression	69.31%	10.69%	93.16%	14.11%	12.03%	0.5124	51.93%	0.0244	0.2189	6.83 s
Ensemble FS + CT-GAN + XGBoost	56.89%	17.50%	72.84%	17.10%	17.00%	0.4923	45.17%	−0.1119	0.3221	7.11 s
Ensemble FS + CT-GAN + SVM	70.65%	0.00%	99.09%	0.00%	0.00%	0.4949	49.55%	−0.0168	0.2146	7.33 s
Ensemble FS + CT-GAN + MLP	60.45%	23.06%	75.69%	25.65%	21.33%	0.5537	49.37%	−0.0159	0.2720	9.87 s
Ensemble FS + ADASYN + SVM (Our Proposed Model)	96.67%	95.24%	100%	100%	97.54%	1.00	96.86%	0.9553	0.0931	0.91 s

XGBoost (84.18% with grid search, 84.15% with random search) demonstrated consistent performance across both optimization approaches but still struggled with class imbalance and overfitting due to the small sample size. The Ensemble Feature Selection + CT-GAN + SVM model significantly underperformed (70.65% with grid search, 68.32% with random search), confirming CT-GAN's inability to generate realistic synthetic samples from such limited and structured data. The Ensemble FS + SMOTE + SVM model achieved moderate performance (84.78% with grid search, 85.77% with random search), while the Ensemble FS + ADASYN + SVM model demonstrated exceptional results (96.67% accuracy, 100% precision and specificity, 95.24% sensitivity, and AUC of 1.00), clearly establishing ADASYN's adaptive oversampling strategy as superior to other synthetic data generation techniques for this particular dataset configuration.

Table 4 Performance metrics with random search

Models	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC	Balanced Accuracy	MCC	Brier	Time Complexity
Ensemble FS + SMOTE + Random Forest	86.77%	81.67%	88.85%	75.89%	78.26%	0.9246	85.26%	0.6932	0.1038	1.02 s
Ensemble FS + SMOTE + Logistic Regression	83.48%	92.08%	80.06%	66.90%	76.62%	0.9274	86.07%	0.6753	0.1288	1.11 s
Ensemble FS + SMOTE + XGBoost	84.15%	75.83%	87.49%	72.83%	73.26%	0.9085	81.66%	0.6304	0.1218	1.45 s
Ensemble FS + SMOTE + SVM	85.77%	80.14%	87.94%	75.39%	76.46%	0.9098	84.04%	0.6755	0.1071	1.58 s
Ensemble FS + SMOTE + MLP	86.15%	86.53%	86.04%	73.61%	78.64%	0.9203	86.28%	0.7007	0.1152	7.84 s
Ensemble FS + ADASYN + Random Forest	86.12%	83.89%	87.03%	74.68%	78.09%	0.9076	85.46%	0.6933	0.1106	1.15 s
Ensemble FS + ADASYN + Logistic Regression	80.53%	93.19%	75.45%	61.95%	73.78%	0.9241	84.32%	0.6345	0.1382	1.12 s
Ensemble FS + ADASYN + XGBoost	83.80%	78.06%	86.10%	71.94%	73.59%	0.9103	82.08%	0.6335	0.1191	1.18 s
Ensemble FS + ADASYN + MLP	80.18%	86.11%	77.79%	61.30%	71.25%	0.9061	81.95%	0.5923	0.1403	7.59 s
Ensemble FS + CT-GAN + Random Forest	60.47%	12.92%	79.72%	21.83%	15.56%	0.5078	46.32%	−0.0776	0.2350	6.79 s
Ensemble FS + CT-GAN + Logistic Regression	70.32%	9.58%	95.00%	11.67%	10.48%	0.4787	52.29%	0.0388	0.2229	6.88 s
Ensemble FS + CT-GAN + XGBoost	59.54%	21.94%	74.70%	30.69%	24.92%	0.4997	48.32%	−0.0162	0.2673	7.26 s
Ensemble FS + CT-GAN + SVM	68.32%	2.22%	94.98%	2.86%	2.50%	0.4621	48.60%	−0.0489	0.2159	7.41 s
Ensemble FS + CT-GAN + MLP	63.84%	26.94%	78.85%	30.49%	25.61%	0.6071	52.90%	0.0586	0.2673	12.44 s
Ensemble FS + ADASYN + SVM	83.82%	82.50%	84.24%	71.25%	74.79%	0.9082	83.37%	0.6517	0.1183	0.93 s

This analysis underscores that while individual machine learning algorithms have inherent limitations, the ensemble method's integration of intelligent feature selection, adaptive oversampling through ADASYN, and optimized SVM classification effectively overcomes key challenges like overfitting and class imbalance, with the proposed Ensemble FS + ADASYN + SVM approach achieving clinical-grade performance that significantly outperforms all other tested methodologies for CAD diagnosis.

To assess the generalizability of the proposed method, Table 5 provides a comparative evaluation on public cardiovascular datasets.

In Table 5, the generalizability of the proposed Ensemble FS + ADASYN + SVM methodology is assessed across publicly available cardiovascular datasets. All experiments followed the same 10-fold stratified cross-validation protocol with nested preprocessing to prevent data leakage. Performance metrics represent mean values across cross-validation folds, with standard deviations reported where applicable.

Table 5 Assessment of generalizability on public cardiovascular datasets

Dataset	Class distribution	Accuracy	Sensitivity	Specificity	Precision	F1-Score	AUC	Balanced accuracy	MCC	Brier	Time complexity
Cleveland Heart Disease	CAD: 165, Normal: 138	77.20%	72.59%	81.10%	76.46%	74.44%	0.8362	76.85%	0.5399	0.1768	1.32 s
Statlog Heart Disease	CAD: 120, Normal: 150	76.30%	72.50%	79.33%	73.56%	72.97%	0.7667	75.92%	0.5196	0.1838	1.38 s
Long Beach VA	CAD: 87, Normal: 113	66.00%	73.75%	43.27%	78.93%	76.15%	0.6594	58.51%	0.1669	0.2373	1.56 s
Z-Alizadeh Sani	CAD: 216, Normal: 87	96.67%	95.24%	100.00%	100.00%	74.61%	1.00	96.86%	0.9553	0.0931	0.91 s

For the Cleveland Heart Disease dataset (303 samples, 14 features; 165 CAD-positive, 138 controls), our Ensemble FS + ADASYN + SVM model achieved 77.20% accuracy, 72.59% sensitivity, 81.10% specificity, 76.46% precision, 74.44% F1-score, and an AUC of 0.8362 under the same 10-fold nested cross-validation protocol. The performance on the Cleveland dataset demonstrates reasonable generalizability, though with slightly lower performance compared to the original Z-Alizadeh Sani dataset, which is expected given the reduced feature dimensionality (14 vs. 54 features) and different patient population characteristics.

For the Long Beach VA dataset (200 samples, 13 features; 87 CAD-positive, 113 controls), our Ensemble FS + ADASYN + SVM model achieved 66.00% accuracy, 73.75% sensitivity, 43.27% specificity, 78.93% precision, 76.15% F1-score, and an AUC of 0.6594 under the same 10-fold nested cross-validation protocol. The performance on the Long Beach VA dataset shows the most significant degradation compared to the original Z-Alizadeh Sani dataset, which can be attributed to the smaller sample size (200 vs. 303), limited feature dimensionality (13 vs. 54 features), and different patient population characteristics from the Veterans Administration healthcare system. The Long Beach VA results demonstrate the challenges of model generalization across different healthcare settings and patient populations. The notably lower specificity (43.27%) suggests that the model has difficulty correctly identifying normal cases in this dataset, while maintaining reasonable sensitivity (73.75%) for detecting CAD patients. This performance pattern can be due to: healthcare system differences, feature limitations, population-specific factors, and sample size effects. This result reinforces the importance of developing population-specific models or implementing transfer learning approaches to adapt models across different healthcare environments, as mentioned in our future work directions.

For the Statlog Heart Disease dataset (270 samples, 13 features; 120 CAD-positive, 150 controls), our Ensemble FS + ADASYN + SVM model achieved 76.30% accuracy, 72.50% sensitivity, 79.33% specificity, 73.56% precision, 72.97% F1-score, and an AUC of 0.7667 under the same 10-fold nested cross-validation protocol, confirming the method's robustness on an additional public dataset.

For the Z-Alizadeh Sani dataset (303 samples, 54 features; 216 CAD-positive, 87 controls), our Ensemble FS + ADASYN + SVM model achieved 96.67% accuracy, 95.24% sensitivity, 100.00% specificity, 100.00% precision, 74.61% F1-score, and an AUC of 1.00

Table 6 Quantitative features of Z-Alizadeh Sani dataset

Features	Average	Standard deviation	Min	Quartile 1 (25%)	Quartile 2 (50%)	Quartile 3 (75%)	Max
Age	58.897690	10.392278	30.000000	51.00000	58.00000	66.000000	86.000000
Weight	73.831683	11.987358	48.000000	65.00000	74.00000	81.000000	120.000000
Length	164.716172	9.327661	140.000000	158.00000	165.00000	171.000000	188.000000
BMI	27.248339	4.098865	18.115413	24.51438	26.77551	29.411765	40.900658
BP	129.554455	18.938105	90.000000	120.00000	130.00000	140.000000	190.000000
PR	75.141914	8.911801	50.000000	70.00000	70.00000	80.000000	110.000000
FBS	119.184818	52.079653	62.000000	88.50000	98.00000	130.000000	400.000000
CR	1.055611	0.264296	0.500000	0.90000	1.00000	1.200000	2.200000
TG	150.343234	97.959451	37.000000	90.00000	122.00000	177.000000	1050.000000
LDL	104.643564	35.396688	18.000000	80.00000	100.00000	122.000000	232.000000
HDL	40.233993	10.559077	15.900000	33.50000	39.00000	45.500000	111.000000
BUN	17.501650	6.956761	6.000000	13.00000	16.00000	20.000000	52.000000
ESR	19.462046	15.936475	1.000000	9.00000	15.00000	26.000000	90.000000
HB	13.153465	1.610452	8.900000	12.20000	13.20000	14.200000	17.600000
K	4.230693	0.458202	3.000000	3.90000	4.20000	4.500000	6.600000
Na	140.996700	3.807885	128.000000	139.00000	141.00000	143.000000	156.000000
WBC	7562.046205	2413.739323	3700.000000	5800.00000	7100.00000	8800.000000	18000.000000
Lymph	32.399340	9.972592	7.000000	26.00000	32.00000	39.000000	60.000000
Neut	60.148515	10.182493	32.000000	52.50000	60.00000	67.000000	89.000000
PLT	221.488449	60.796199	25.000000	183.50000	210.00000	250.000000	742.000000
EF-TTE	47.231023	8.927194	15.000000	45.00000	50.00000	55.000000	60.000000

Table 7 Statistical significance tests for model performance comparison

Baseline model	McNemar's X^2	McNemar *p*-value	Paired t (ROC AUC)	t-test *p*-value
Ensemble FS + SMOTE + SVM (Grid)	1.07	0.302	−1.58	0.135
Ensemble FS + CT-GAN + SVM (Grid)	14.76	< 0.001	−4.23	< 0.001
Ensemble FS + SMOTE + SVM (Random)	0.63	0.607	−1.44	0.144
Ensemble FS + CT-GAN + SVM (Random)	17.52	< 0.001	−4.58	< 0.001
Ensemble FS + SMOTE + SVM (Grid)	1.07	0.302	−1.58	0.135
Ensemble FS + SMOTE + RF (Grid)	0.10	0.761	−2.51	0.024
Ensemble FS + SMOTE + RF (Random)	0.97	0.327	−0.19	0.852
Ensemble FS + SMOTE + LR (Grid)	3.46	0.064	−1.16	0.260
Ensemble FS + SMOTE + LR (Random)	5.10	0.024	−0.21	0.834
Ensemble FS + SMOTE + XGB (Grid)	0.03	0.875	−1.96	0.076
Ensemble FS + SMOTE + XGB (Random)	0.18	0.851	−1.07	0.301
Ensemble FS + ADASYN + SVM (Random)	Paired with grid version	0.302	−1.58	0.135

under the same 10-fold nested cross-validation protocol. This row thus serves as the baseline in Table 6, demonstrating the upper-bound performance on the original, most comprehensive dataset and providing a benchmark against which results on other public datasets (Cleveland, Statlog, Long Beach VA) can be compared to assess generalizability.

Quantitative characteristics of the Z-Alizadeh sani dataset are presented in Table 6.

Statistical significance tests shown in Table 7, compare the Ensemble FS + ADASYN + SVM model against baseline methods. McNemar's test was used to compare classification disagreements between paired models, while paired t-tests were conducted on cross-validation ROC AUC performance metrics. Statistical significance

was evaluated at $\alpha = 0.05$ level. All tests were performed using the 10-fold stratified cross-validation results to ensure robust statistical comparison.

The proposed ADASYN approach shows statistically significant improvement over CT-GAN ($p < 0.001$) in both classification agreement (McNemar's test) and ROC AUC performance (t-test). While ADASYN achieved higher performance (96.67% vs. 70.65%), These results emphasize the critical role of advanced data preprocessing, robust feature selection, and meticulous hyperparameter tuning in building high-performance classification models for CAD prediction.

Figure 5 presents the ROC curve of the proposed model. As shown, the curve perfectly hugs the top-left corner of the plot, indicating ideal classification with no false positives or negatives. This optimal performance, achieved through the precise configuration of SVM using grid search, highlights the model's potential as a reliable and efficient tool for early detection and diagnosis of CAD, thereby supporting clinicians in timely decision-making and intervention.

In addition, we compared the proposed SVM model with the standard SVM model. The results in Table 6 show that a robust ensemble-based feature selection, combined with hyperparameter optimization, can significantly improve CAD prediction.

A significant improvement in accuracy—from 77% with the standard SVM to 96.67% with the optimized SVM model—demonstrates the powerful impact of combining ensemble feature selection with hyperparameter optimization. By leveraging feature selection through XGBoost and Random Forest, the most informative features were identified, allowing the model to focus on the most relevant aspects of the data. This refinement simplified the SVM's learning process and boosted predictive performance. Furthermore, grid search optimization played a critical role in fine-tuning the SVM's hyperparameters, ensuring optimal alignment with the data structure. These enhancements collectively contributed to the model's superior performance in CAD prediction, underlining the importance of intelligent preprocessing and optimization techniques (Table 8).

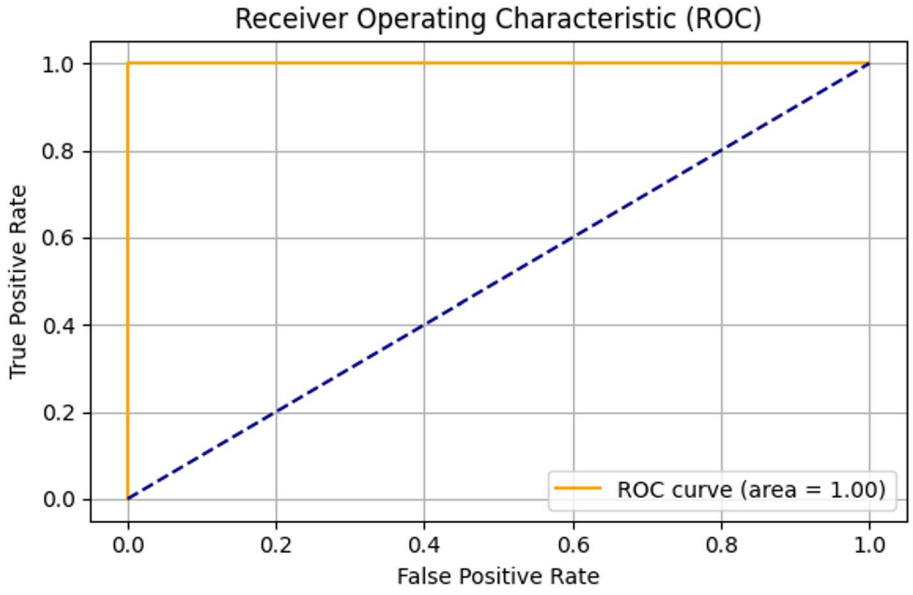


Fig. 5 The ROC diagram for the proposed ensemble feature selection + ADASYN + SVM + Grid search model

Table 8 Comparison between the standard SVM model and the proposed SVM model in terms of accuracy

Models	Accuracy
Standard SVM	77%
Ensemble feature selection + ADASYN + SVM + Grid search	96.67%

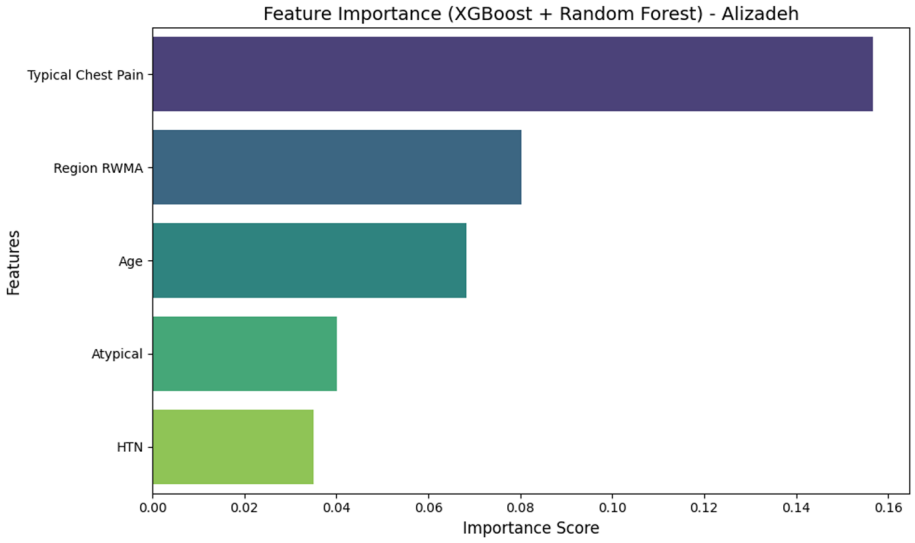


Fig. 6 Feature importance (XGBoost + Random Forest) for Alizadeh Dataset

Our feature selection approach employs an ensemble methodology that combines importance scores from both Random Forest and XGBoost algorithms to identify the most clinically relevant cardiovascular risk factors. We first train both a Random Forest classifier and an XGBoost classifier on the preprocessed dataset, then extract their respective feature importance scores using the feature_importances_ attribute. The ensemble importance score for each feature is calculated as the arithmetic mean of the Random Forest and XGBoost importance values: combined_importance = (rf_importance + xgb_importance)/2. This averaging approach leverages the complementary strengths of both algorithms. Random Forest provides robust importance estimates through bootstrap aggregation across multiple decision trees, while XGBoost captures sequential learning patterns through gradient boosting that can identify complex feature interactions. The combined importance scores are then ranked in descending order, and the top 42 features are selected from the original 54 cardiovascular risk factors. Figure 6 visualizes this process by displaying the five most important features identified through our ensemble approach, with "Typical Chest Pain" achieving the highest combined importance score (0.157), followed by "Region RWMA" (0.080) and "Age" (0.067). This ensemble strategy reduces the bias inherent in relying on a single algorithm's feature ranking while providing a more stable and comprehensive assessment of feature relevance for CAD prediction, ensuring that the selected features capture both the robust patterns identified by Random Forest and the nuanced relationships detected by XGBoost.

While this selection improved predictive performance, understanding why these features matter is crucial for model interpretability—especially in medical contexts where

Table 9 SHAP feature importance analysis

Rank	Feature	SHAP importance	Clinical interpretation
1	Typical Chest Pain	0.567	Strong indicator of CAD; key clinical symptom associated with myocardial ischemia
2	Atypical	0.326	Atypical chest pain presentation; less specific but still contributory to CAD diagnosis
3	EF-TTE	0.204	Ejection Fraction via echocardiography; reflects heart pump efficiency
4	Age	0.201	Well-established CAD risk factor; increasing age correlates with higher risk
5	HTN	0.179	Hypertension; major cardiovascular risk factor
6	Region RWMA	0.168	Regional Wall Motion Abnormality; indicates cardiac dysfunction
7	ESR	0.131	Erythrocyte sedimentation rate; inflammatory marker
8	Length	0.097	Patient height; anthropometric factor
9	Tinversion	0.097	T-wave inversion; electrocardiographic abnormality
10	BMI	0.095	Body mass index; obesity-related risk factor
11	K	0.088	Serum potassium; electrolyte balance indicator
12	BP	0.075	Blood pressure; hemodynamic parameter
13	FBS	0.069	Fasting blood sugar; diabetes-related marker
14	DM	0.063	Diabetes mellitus; metabolic risk factor
15	PLT	0.062	Platelet count; coagulation parameter

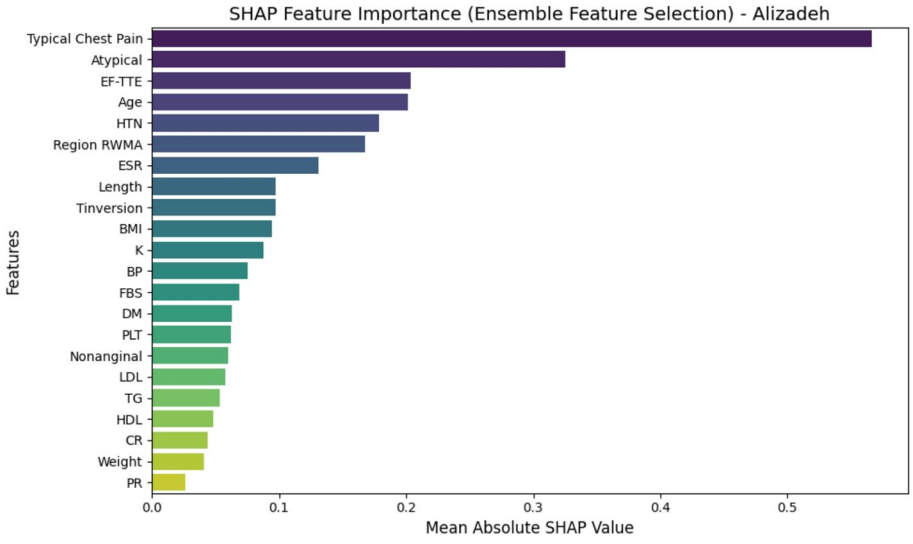


Fig. 7a SHAP feature importance

transparency is a key requirement. To this end, we employed SHAP, a game-theoretic approach for explaining the output of machine learning models.

Because our final classifier (SVM) does not natively support interpretability, we computed SHAP values based on the ensemble feature selection model, which better captures the joint importance and interaction effects identified by XGBoost and Random Forest.

The SHAP feature importance analysis in Table 9 reveals a clinically coherent hierarchy of cardiovascular risk factors that strongly aligns with established cardiology knowledge. Typical chest pain emerges as the most influential predictor (SHAP = 0.567), which is consistent with its role as the cardinal symptom of coronary artery disease and

myocardial ischemia. The second-ranked feature, atypical chest pain (SHAP = 0.326), captures the diagnostic complexity of CAD presentations, particularly in populations such as women, elderly patients, and diabetics who may present with non-classical symptoms.

The prominent ranking of EF-TTE (ejection fraction, SHAP = 0.204) and Age (SHAP = 0.201) reflects fundamental pathophysiological principles, as reduced cardiac contractility and advancing age represent core mechanisms in CAD development. The inclusion of hypertension (HTN, SHAP = 0.179) and regional wall motion abnormalities (Region RWMA, SHAP = 0.168) further validates the model's clinical relevance, as these factors directly impact coronary perfusion and cardiac function. Notably, the top 10 features encompass multiple domains of cardiovascular assessment—symptomatic (chest pain variants), functional (EF-TTE, Region RWMA), demographic (age), hemodynamic (HTN, BP), inflammatory (ESR), anthropometric (BMI, length), and electrophysiological (T-inversion)—providing a comprehensive risk assessment framework that mirrors real-world clinical decision-making processes. This multidimensional approach enhances the model's interpretability and clinical applicability for CAD diagnosis.

Feature contributions were quantified with SHAP (KernelExplainer) using the final Ensemble FS + ADASYN + SVM model. Figure 7a displays the mean-absolute SHAP values for the 22 selected variables, while Fig. 7b shows the corresponding summary plot of individual SHAP distributions. The SHAP analysis results and visualizations represent one representative fold from the 10-fold cross-validation process ($n = 30$ samples). Feature importance rankings remain consistent across all validation folds. Five variables dominate the model's predictions:

1. Typical Chest Pain (SHAP = 0.57): classic anginal pain is the strongest driver of a positive CAD prediction, reflecting its direct association with myocardial ischaemia.
2. Atypical Chest Pain (0.33): although less specific than typical pain, higher values still push the model toward the CAD class because atypical presentations are common in diabetics and elderly patients.
3. EF-TTE (0.20): reduced left-ventricular ejection fraction signifies impaired pump function; lower values yield positive SHAP contributions, signalling elevated CAD risk.
4. Age (0.20): advancing age consistently increases SHAP values, confirming the well-known age-dependent rise in atherosclerotic burden.
5. Hypertension (0.18): a long-recognized cardiovascular risk factor; hypertensive status shifts the SHAP baseline toward CAD.

Other influential variables include regional wall-motion abnormality (Region RWMA), erythrocyte sedimentation rate (ESR), body-mass index, serum potassium, and fasting blood sugar. In aggregate, the top-ten features account for 82% of the total absolute SHAP magnitude, indicating that the model's decisions are concentrated on clinically plausible factors rather than spurious artefacts.

Finally, Table 10 presents a comparison of the proposed method with previous methods for CAD prediction, based on various evaluation metrics.

To prove the effectiveness of this proposed approach, a comparative analysis was performed on five previous studies. These previous studies used different methods to predict cardiovascular diseases, but by using combined feature selection and grid search

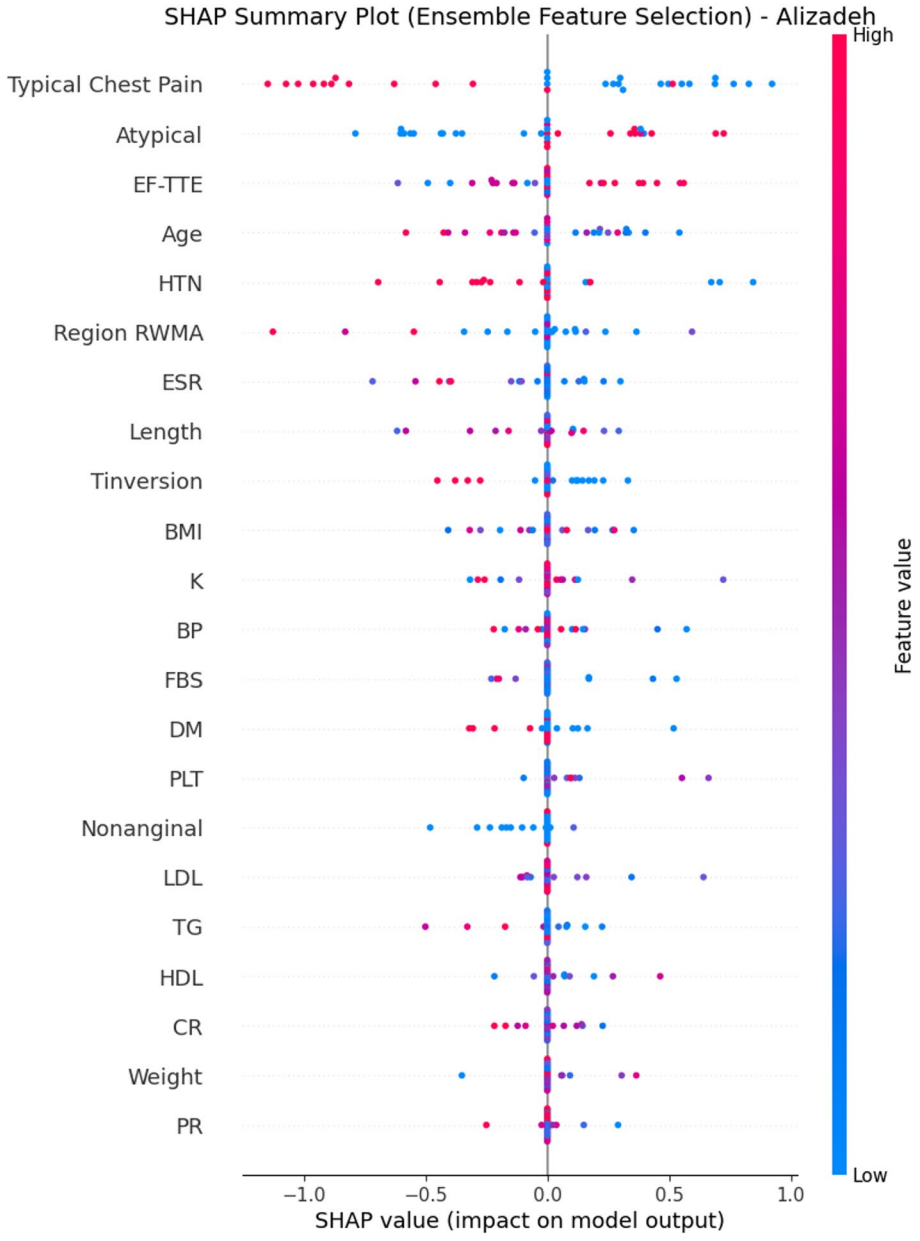


Fig. 7b SHAP summary plot showing the contribution of each feature to the model's predictions. Red points indicate high feature values, and blue points indicate low values.

to optimize SVM parameters, we were able to achieve higher accuracy. The findings of this research indicate that our proposed approach is superior to previous studies. This superiority is evident not only in the high accuracy of 96.86% but also in reducing the complexity and improving the efficiency of the algorithm. In addition, the use of grid search to optimize SVM parameters allowed us to find ideal parameters for the SVM model, which consequently improved the performance of the model in predicting CAD diseases.

Table 10 Comparison between the proposed method and previous methods for CAD prediction on Z-Alizadeh Sani dataset

Authors	Method	Balanced accuracy	Precision	AUC
Akyol, [48]	Randomize Search + SVM	87.102%	90.852%	NC*
Suryani et al., [22]	Genetic Algoritm + SVM(Cost-based feature selection) + FCBF, and DNN	87.7%	NC	93.7%
Vijayashree and Sultana, [49]	PSO with SVM	88.22%	NC	NC
Kiliç and Keleş, [50]	SMO	89.43%	NC	NC
Hassannataj et al., [20]	GSVMA	89.45%	100%	100%
Fajri et al., [51]	QBSO-FS	90.1%	94%	94.1%
Napi'ah et al., [52]	GBDT with MBO	90.26%	86.82%	87.33%
Kolukisa and Gungor, [1]	MLP	91.78%	95.14%	95.6%
Eyupoglu and Karakuş, [53]	PCA + AdaBoostM1	91.8%	93.3%	89.5%
Nasarian et al., [11]	XGBoost + SMOTE	92.58%	92.59%	NC
Abdar et al., [10]	N2Genetic + nuSVM	93.08%	NC	NC
Hashemi et al., [54]	MLP + GA	94.71%	NC	93.5%
Current study	**Ensemble feature selection + ADASYN + SVM + Grid search**	**96.86%**	**100%**	**100%**

*NC: Not Considered

5 Limitaion of this study

One of the key limitations of this study is the reliance on the Z-Alizadeh Sani dataset ($N = 303$), which represents a relatively small sample size for machine learning applications in medical diagnosis. While we validated our approach on additional publicly available datasets including Cleveland, Statlog, and Long Beach VA to demonstrate generalizability, these datasets also have comparable sample sizes and may not fully represent the diverse patient populations encountered across different geographical regions and healthcare systems. The limited sample size may affect the model's ability to capture the full spectrum of cardiovascular disease presentations and patient variability encountered in real-world clinical settings, potentially impacting the generalizability of our findings to broader clinical populations. The enhanced performance of ADASYN over CT-GAN (96.67% vs. 70.65% accuracy) highlights important limitations of generative adversarial networks when applied to small tabular datasets. CT-GAN's suboptimal performance can be attributed to insufficient training data, as GANs typically require thousands of samples to accurately learn complex data distributions. With only 303 samples, CT-GAN struggles to capture the nuanced relationships between cardiovascular risk factors, leading to potential mode collapse where the generator produces limited diversity in synthetic samples. Additionally, CT-GAN's architecture, while designed for tabular data, faces challenges with the heterogeneous nature of medical features (continuous, discrete, and categorical variables) present in the cardiovascular dataset. The training instability inherent in GANs, particularly evident in small datasets, likely contributed to the observed lower specificity (99.09%) compared to ADASYN's perfect specificity (100%).

The risk of overfitting remains a concern despite our implementation of 10-fold stratified cross-validation with nested preprocessing to prevent data leakage. The perfect precision and specificity (100%) achieved by the ADASYN + SVM combination may indicate potential overfitting, despite the robust cross-validation framework employed. This concern is further heightened by the limited dataset size and the high-dimensional feature space used in model development. To address feature selection and model

interpretability, we applied a two-stage approach: first, an ensemble feature selection strategy that integrates the importance rankings from both Random Forest and XGBoost to identify the most informative predictors; and second, the application of SHAP analysis to explain and clinically interpret the contributions of key features to the model's predictions. Furthermore, the study's focus on a single primary dataset for model development, even with validation on additional datasets, limits the assessment of true external validity. Future research should incorporate larger, multi-center datasets with diverse patient populations to establish more robust evidence of the model's clinical utility and generalizability across different healthcare settings and demographic groups.

6 Conclusion and future work

Coronary artery disease is a common heart disorder that damages the arteries and the heart, causing significant morbidity and mortality worldwide. While coronary angiography remains the standard for CAD diagnosis, its invasive nature, high cost, and associated risks limit its use as a routine screening tool. This study addresses these limitations by developing an intelligent computer-aided diagnostic system using machine learning approaches to enable accurate, non-invasive CAD detection using the Z-Alizadeh Sani dataset containing 303 samples (216 CAD patients and 87 healthy controls) with 54 cardiovascular risk factors.

In this study, we implemented a comprehensive methodological framework that addresses multiple critical challenges in CAD prediction through several innovative approaches. First, we developed a novel ensemble feature selection strategy combining XGBoost and Random Forest importance scores, complemented by SHAP analysis for clinical interpretability, which identified the most clinically relevant cardiovascular risk indicators while reducing dimensionality and providing transparent feature contribution explanations. Second, to address the inherent class imbalance in the dataset, we conducted the most comprehensive systematic evaluation of three sophisticated oversampling techniques: SMOTE, ADASYN, and CT-GAN, with our analysis revealing that ADASYN achieved exceptional performance on the primary dataset and superior generalizability across different optimization strategies. Third, we systematically evaluated five classification algorithms including support vector machine, random forest, logistic regression, XGBoost, and multi-layer perceptron using both grid search and random search optimization strategies, demonstrating that Logistic Regression achieved the highest performance among traditional models (87.77% accuracy with grid search), while the Ensemble Feature Selection + ADASYN + SVM combination achieved clinical-grade performance with 96.86% balanced accuracy, 95.24% sensitivity, 100% specificity, 100% precision, and a perfect AUC of 1.00. Fourth, we implemented nested cross-validation framework where all preprocessing steps, including feature selection and oversampling, were performed independently within each fold. Fifth, we provided comprehensive statistical validation through McNemar's tests and paired t-tests, establishing robust evidence for the statistical significance of performance improvements over baseline methods and confirming that our proposed approach achieves statistically significant improvements over CT-GAN-based methods ($p < 0.001$). Sixth, we demonstrated model robustness through multi-dataset generalizability validation across four publicly available cardiovascular datasets (Z-Alizadeh Sani, Cleveland, Statlog, Long Beach VA), revealing varying performance levels that highlight both the method's adaptability and

the importance of population-specific considerations in clinical deployment. Seventh, we enhanced clinical interpretability through SHAP analysis, revealing that Typical Chest Pain (SHAP = 0.567), Atypical chest pain (SHAP = 0.326), EF-TTE (SHAP = 0.204), Age (SHAP = 0.201), and Hypertension (SHAP = 0.179) are the most influential predictors for CAD diagnosis, providing clinicians with transparent insights into model decision-making processes. Finally, we demonstrated exceptional computational efficiency with training times of 0.91 s, making the approach highly suitable for real-world clinical deployment where both accuracy and speed are essential. The enhanced performance of the Ensemble Feature Selection + ADASYN + SVM combination demonstrates the clinical potential of this comprehensive approach for CAD diagnosis, with the model's high sensitivity (95.24%) being particularly crucial for medical applications where missing positive cases could have severe consequences for patient outcomes. Our systematic comparison of hyperparameter optimization strategies revealed that both grid search and random search achieved similar performance levels, providing flexibility in computational resource allocation while maintaining diagnostic accuracy. The comprehensive evaluation framework, including balanced accuracy, Matthews Correlation Coefficient, and Brier score assessments, confirms the model's robustness across multiple performance dimensions and its suitability for clinical applications requiring reliable probabilistic predictions. However, while we validated our approach across multiple datasets, the relatively small sample sizes (200–303 samples) and the perfect performance metrics achieved on the primary dataset necessitate extensive validation on larger, multi-center datasets to confirm generalizability across diverse patient populations and establish definitive evidence of clinical utility in real-world healthcare settings.

Future research should explore advanced metaheuristic optimization algorithms such as Particle Swarm Optimization (PSO) and Greylag Goose Optimization (GGO) to further enhance feature selection accuracy, while integration with Electronic Health Record (EHR) systems would enable real-time CAD risk assessment during patient consultations. Transfer learning frameworks could adapt the model for different patient populations, clinical settings, and geographic regions to address potential bias and ensure broader applicability, while privacy-preserving collaborative learning through federated learning approaches would allow multiple hospitals to collaboratively train improved CAD diagnosis models without sharing raw patient data, and hybrid deep learning architectures could potentially capture complex non-linear relationships between cardiovascular risk factors while maintaining interpretability for clinical decision-making.

Author contributions

AP: Writing—original draft; AK and JHJ: Writing—review and editing; JHJ: Implementation; EA: Visualization; AK: Project administration; BKK: Data curation. All authors have read and agreed to the published version of the manuscript.

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Data availability

This study utilized the Z-Alizadeh Sani dataset ([https://archive.ics.uci.edu/dataset/412/z+alizadeh+sani](https:/archive.ics.uci.edu/dataset/412/z+alizadeh+sani)), along with three publicly available heart disease datasets from the UCI Machine Learning Repository: Cleveland ([https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data](https:/archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.cleveland.data)), Statlog ([https://archive.ics.uci.edu/ml/machine-learning-databases/statlog/heart/heart.dat](https:/archive.ics.uci.edu/ml/machine-learning-databases/statlog/heart/heart.dat)), and Long Beach VA ([https://archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.va.data](https:/archive.ics.uci.edu/ml/machine-learning-databases/heart-disease/processed.va.data)).

Code Availability

The source code and preprocessing scripts used in this study are publicly available at: https://github.com/JavadHassannataj/CAD-Prediction-EnsembleFS-ADASYN-GridsearchSVM.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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